SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 5

Strategic Capacity Planning

Strategic capacity planning (Ch. 5) plus strategy lenses (RBV/VRIN and dynamic capabilities) and the Pelindo & Semarang Container Terminal case. Covered: capacity as a strategic commitment; capacity, best operating level and utilisation; economies and diseconomies of scale; focus and flexibility; step size and timing; determining requirements (forecast → convert → project → cushion → chunk → balance); decision trees (Hackers, undiscounted and at 16%); service capacity and utilisation zones; RBV & VRIN; Five Forces → RBV → dynamic capabilities; and six TPKS discussion questions.

Material

1 Learning objectives and roadmap

1. Explain capacity management and why the level is a strategic commitment.
2. Plan capacity: requirements, cushions, chunk size, system balance.
3. Evaluate alternatives with decision trees.
4. Compare service and manufacturing capacity through RBV and dynamic capabilities.

2 Why capacity is a strategic problem

- Capacity is the output a system can achieve over a period — a rate.
- Strategic capacity planning sets capital-intensive resources for long-term strategy.
- Horizons: long (>1 yr), intermediate (6–18 months), short (<1 month).
- Measure in outputs if the mix is stable, inputs if not.

Lead time
Bought against a forecast.

Irreversibility
Shedding means selling or liquidating.

Asymmetry
Too little loses customers; too much forces discounting.

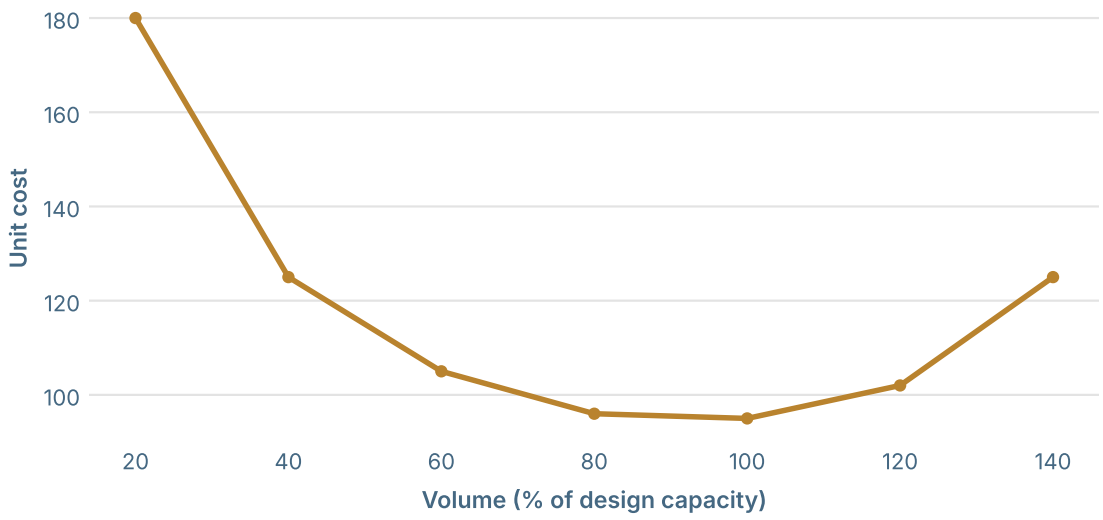
3 Capacity, best operating level and utilisation

Term	Definition
Capacity	Achievable output over a period — always state the period.
Best operating level	Volume minimising average unit cost.
Utilisation rate	Capacity used ÷ best operating level.

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Utilisation = used ÷ best operating level → 480 ÷ 500 = 96%
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Below: fixed cost spread thin; at: lowest cost but no cushion; above: overtime, wear, defects — and queues in services.

Average unit cost vs volume (conceptual)



4 Economies and diseconomies of scale

1. **Economies of scale** — lower cost per unit as volume rises.
2. **Turning point** — keeping a huge facility busy may need discounting; an M&M line moves 2.6 million sweets an hour with heavy maintenance labour.
3. **Diseconomies** — coordination and complexity push cost up again.

Plant size
Often set by transport cost (a cement plant serves a local market).

! **Scale ≠ productivity**
Jaguar ~15 cars per employee; Volvo Torslanda 29; BMW Mini 39 with 4,500 workers.

5 Capacity focus and capacity flexibility

A **focused factory** pursues a limited set of objectives. Ways: plant within a plant; flexible plants (near-zero changeover); flexible processes (economies of scope); flexible workers.

PWP zone	Process	Focus
High volume, low variety	Flow line	Cost
Low volume, high variety	Job shop	Flexibility
New product ramp	Pilot cell	Learning
Service & spares	Fast response	Delivery

Economies of scope; several best levels under one roof; flexibility is bought up front.

6 Determining requirements, cushions and chunks

1. Forecast
2. Convert to equipment and labour
3. Project against availability
4. Cushion
5. Chunk
6. Balance

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Capacity cushion = (capacity − demand) ÷ demand
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Example

Demand $10m, capacity $12m → 20% cushion = 83% utilisation; demand $12m vs capacity $10m → −16.7%.

Trade-off	Frequent small	Infrequent large
Idle capacity cost	Low	High
Shortage risk	Higher	Lower
Scale economies	Forgone	Captured
Disruption	Repeated	One-off
Capital	Staged	Front-loaded

7 Considerations in changing capacity

Consideration	Explanation
System balance	Add at the bottleneck, buffer in front of it, or duplicate.

Frequency	Too often = repeated disruption; too rarely = large chunks as overhead.
External sources	Outsourcing; sharing (airlines swap aircraft).
Decreasing	Harder than adding; permanent cuts mean selling or liquidating.

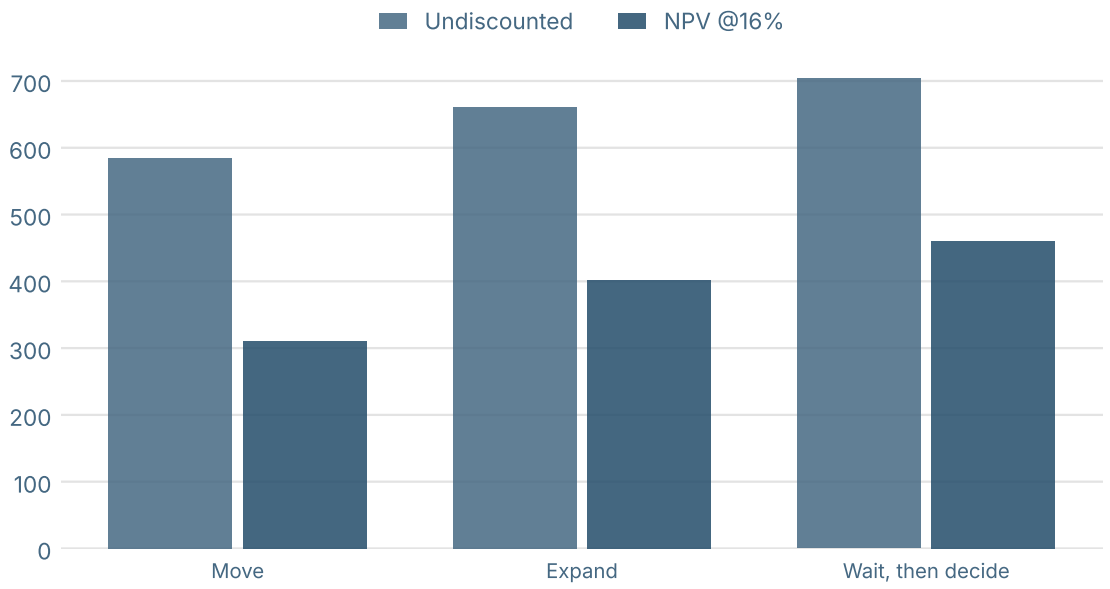
8 Decision trees: Hackers Computer Store

Squares are decisions, circles are chance events. Hackers (5 years, 55% strong growth):

Option	Cost	Per year (strong / weak)	EV
Move	$210,000	$195,000 / $115,000	$585,000
Expand	$87,000	$190,000 / $100,000	$660,500
Do nothing (option to expand in year 2)	$0	$170,000 / $105,000	$703,750

Solve right to left; the tree also says don't expand in year two. At 16%: $310,613, $402,507, $460,857 (−47%, −39%, −35%) — ranking holds, margins compress, timing matters.

Hackers: expected values ($000)

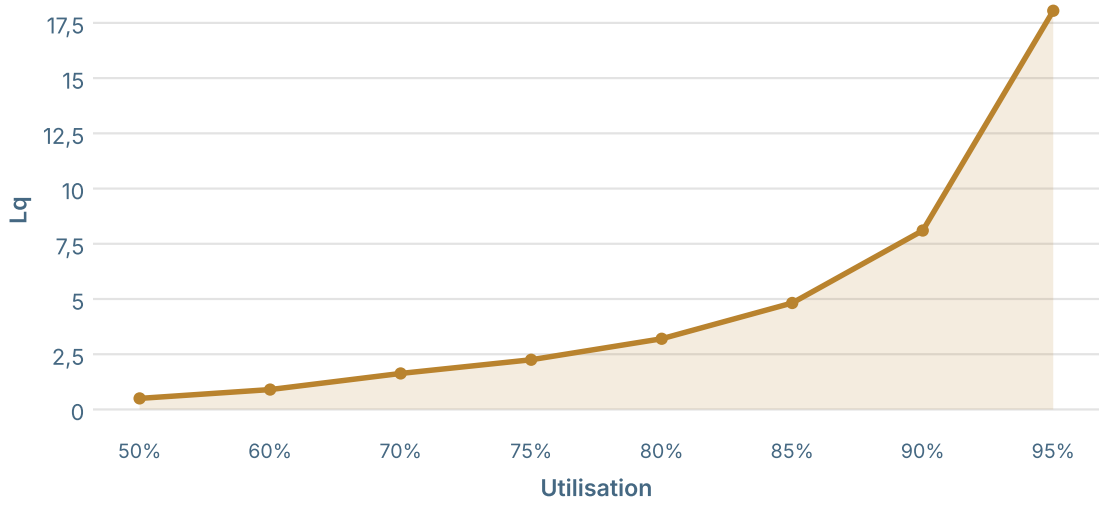


9 Service capacity is a different problem

Dimension	Manufacturing	Services
Storability	Inventory buffers	Capacity perishes hourly
Location	Produce centrally, distribute	Distribute capacity first
Volatility	Smoothed	Higher
Planning increment	Weekly/monthly	10–30 minutes
Utilisation target	Near 100%	~70%
Running full	Unit cost rises	Queue and quality fall first
Lever	Build, subcontract	Reprice peaks, reshape demand, extend hours

A service at 47% annual utilisation can still turn customers away at 9 a.m. Zones: idle, service (best ~70%), critical, non-service.

Queue length vs utilisation (single server)



Low targets for ERs; ~70% for most services; near 100% for commuter trains; full for stadiums.

10 The resource-based view and the VRIN test

Test	Question	If no
Valuable	Exploit opportunity / neutralise threat?	Disadvantage
Rare	Few competitors hold it?	Parity
Inimitable	Costly to copy?	Temporary advantage
Non-substitutable	No equivalent route?	Temporary advantage
Organised	Structured to capture value?	Unexploited potential

Sequential test. Capacity anyone can buy is valuable but not rare → parity; resources that pass usually constrain capacity (site, right-of-way, licence, relationship).

11 From Five Forces to dynamic capabilities: sensing, seizing, transforming

Framework	Advantage	Capacity question
Five Forces (outside-in)	Industry structure	Will the industry pay for the capacity?
RBV (inside-out)	VRIN assets	Is it rare and hard to copy?
Dynamic capabilities	Reconfiguring the asset base	Can we reconfigure if wrong?

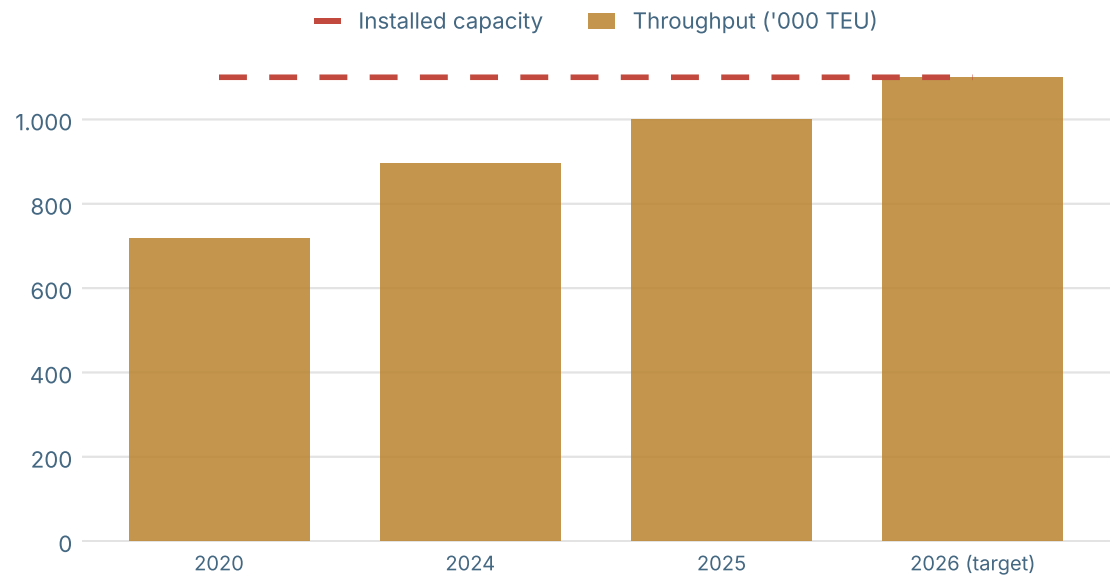
Teece: ordinary capabilities (do things right, imitable); **sensing** (scan, interpret), **seizing** (business model, alliances, investment protocols), **transforming** (restructure, redeploy, divest) — a continuous cycle.

Capacity as orchestration: never neutral; sunk-cost rule; binding constraint may be land, licence or relationship; conversion often beats construction. Moves: sense before you size, seize in defensible chunks, transform before you duplicate.

12 Pelindo & Semarang Container Terminal: adding capacity on sinking ground

September 2026: Andira Prawitasari has three weeks before the 2027–2031 capital proposal. TPKS ~1m TEU (2025), installed 1.1m TEU = 2026 target; ~70% of Central Java boxes ship via Priok and Perak.

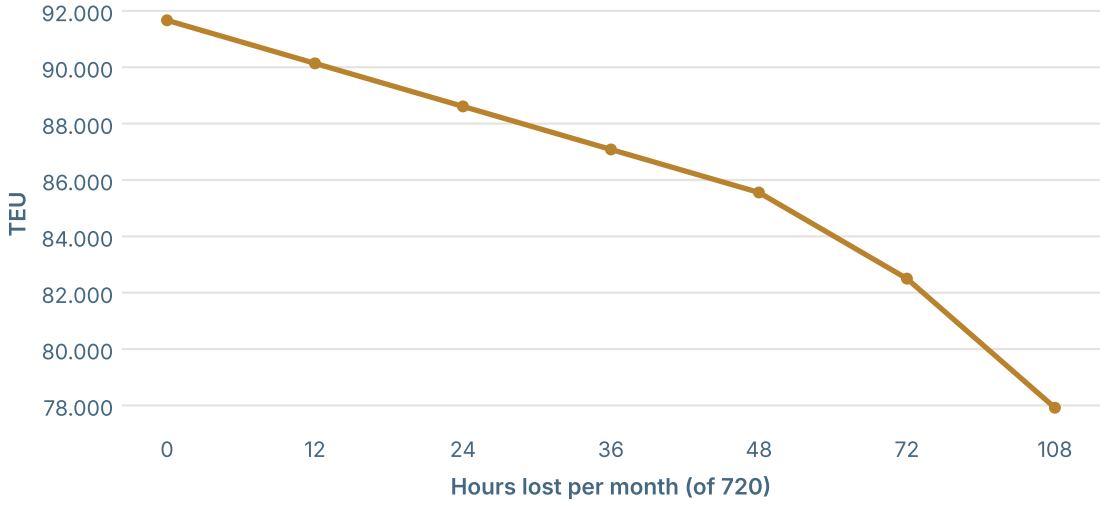
TPKS throughput ('000 TEU) and utilisation



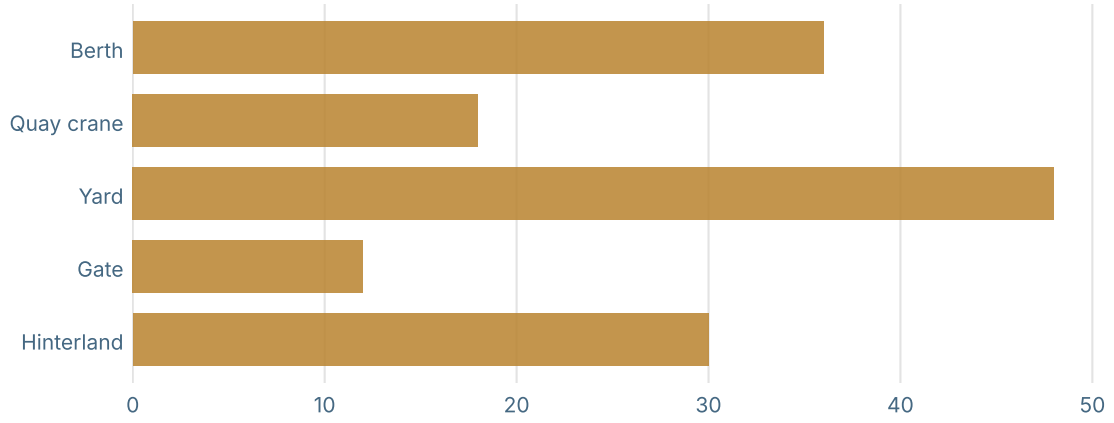
Utilisation 65.2% → 81.4% → ~90.9% → 100%.

Demand: 717,062 TEU (2020) → 895,904 (2024) → >1m (2025); 1.7m circulating for 2027 vs an official 1.2m by 2029. Ground: subsidence of a dozen-plus cm a year; May 2025 sea-wall collapse (~25 m), ~75% of the port flooded, berth halted. Demand moving west: Kendal 100% taken up, KITB Batang ~60%; two ~50 ha dry-port candidates.

Effective monthly capacity vs flood hours lost



Investment lead time by stage (months, midpoint)



Midpoints of the case ranges.

! **Binding question**

Is TPKS really the binding stage, or is it ground elevation and hinterland choice?

Discussion: state the period; add at the bottleneck (yard, outside-the-fence constraints); bring subsidence in as net available operating hours; decision tree with irreversible branches behind triggers; VRIN — the permitted site is rare precisely because it can't move; design a routine with a trigger number, small reversible commitments and open options.

13 Debrief: six things from the Pelindo case

1. State the period out loud.
2. Cushion is a decision.
3. Find the bottleneck before sizing the asset.
4. The binding constraint may not be machine-shaped.
5. Chunk and timing are the real decision — on sinking ground, too large costs more than too small.
6. The source of advantage can be the limit.

Key takeaways

1. Capacity is a rate over a stated period; three horizons.
2. Lead time, irreversibility, asymmetry.
3. Best operating level and utilisation (480 ÷ 500 = 96%).
4. Economies → diseconomies; transport sets plant size; scale ≠ productivity.
5. Focused factory, PWP, flexibility, economies of scope.
6. Forecast → convert → project → cushion → chunk → balance.
7. Balance, frequency, external sources, decreasing.
8. Decision trees: Hackers EVs and 16% NPVs.
9. Service capacity and the ~70% target.
10. VRIN(O).
11. Five Forces → RBV → dynamic capabilities.
12. TPKS lessons.

Case studies

Case & worked answers

Pelindo & Semarang Container Terminal: capacity on sinking ground

Throughput grew to >1m TEU on 1.1m capacity, giving a zero 2026 cushion; subsidence >12 cm/yr and a May 2025 flood halted operations. Q1: annual utilisation hides peaks — measure weekly YOR, dwell time, vessel waits. Q2: the yard is the least elastic stage; cranes alone move the queue ashore; the hinterland may bind. Q3: effective capacity = installed × net available operating hours index (720 − 36)/720 = 95% → ~87,083 TEU/month. Q4: full B is irreversible and risks stranding; the hybrid has the best EV/downside. Q5: the Tanjung Emas site is VRIN yet caps capacity. Your syndicate's recommendation: sequential hybrid with Chunk 1 (US$8–12m), Chunk 2 dry-port JV (US$20–25m),

YOR ≥85% trigger and subsidence ≥15 cm or downtime ≥10% kill criteria. Improvement: quantify probabilities and EVs.

Worked examples

Problem 1 Easy **Utilisasi dan cushion**

Lini produksi: best operating level 100 unit/jam, output aktual 85 unit/jam. Pabrik lain: kapasitas 1.200 unit/hari, permintaan 1.000 unit/hari. Hitung utilisasi lini pertama dan capacity cushion pabrik kedua.

1. `Utilisasi = Kapasitas terpakai ÷ Best operating level = 85 ÷ 100 = 85%`

2. `Cushion = (Kapasitas − Permintaan) ÷ Permintaan = 200 ÷ 1.000 = 20%`

Answer: Utilisasi 85%; cushion 20%.

Problem 2 Medium **Kapasitas efektif saat banjir**

Kapasitas terpasang 100.000 TEU/bulan pada 720 jam terjadwal. Bulan ini 54 jam hilang karena banjir rob. Berapa kapasitas efektif?

1. Index = (720 − 54) ÷ 720 = 0,925.

2. Kapasitas efektif = 100.000 × 0,925 = 92.500 TEU.

Answer: 92.500 TEU (hilang 7.500 TEU).

Problem 3 Medium **Decision tree ekspansi**

Permintaan tinggi p = 0,6, rendah 0,4. Ekspansi besar: 500 / −100. Ekspansi kecil: 300 / 100 (Rp juta). Mana yang dipilih?

1. EV besar = 0,6 × 500 + 0,4 × (−100) = 300 − 40 = 260.
2. EV kecil = 0,6 × 300 + 0,4 × 100 = 180 + 40 = 220.
3. Besar unggul 40, tetapi berisiko rugi 100 jika permintaan rendah.

Answer: Ekspansi besar (EV 260 vs 220) — kecuali perusahaan tidak sanggup menanggung kemungkinan rugi Rp100 juta.

Practice questions

Try answering first. The answer key is below each question.

1. Capacity 1.1m TEU, throughput 1.0m. Utilisation…

A. 81%
B. 90.9%
C. 100%
D. 110%

Answer: B. 1.0/1.1.

2. Adding cranes when the yard is full…

A. Lifts throughput
B. Moves the queue ashore
C. Cuts dwell time
D. Reduces floods

Answer: B. Add at the bottleneck.

3. Face-to-face services typically target about…

A. 40%
B. 70%
C. 95%
D. 100%

Answer: B. Above that, queues grow.

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4. A resource anyone can buy (cranes) gives, per VRIN…

A. Sustained advantage
B. Parity
C. Temporary advantage
D. Disadvantage

Answer: B. Valuable but not rare.

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5. 'Transforming' is best illustrated by…

A. Building a duplicate plant
B. Converting an underused gas line to oil
C. Overtime
D. Raising prices

Answer: B. Energy East.

Sources

- P3MD MK4 OSCM — Strategic Capacity Planning (class slides)
- Jacobs & Chase — OSCM Ch. 5; Barney (1991); Teece (1997, 2007); Pelindo & TPKS case